

ENTERPRISE HOME LAB

Kali Linux Deployment and System Maintenance

NGT appliance import, network validation, rolling updates, cleanup, and operational verification

PORTFOLIO DOCUMENTATION

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Executive Summary

This guide documents the deployment and maintenance of an NGT-provided Kali Linux virtual appliance in Oracle VirtualBox. The project includes appliance-source documentation, import configuration, dual-network validation, package updates, system cleanup, and a final operational health check.

Deployment Outcome

Kali Linux was imported successfully, updated to Kali GNU/Linux Rolling 2026.1, validated on kernel 7.0.12+kali-amd64, and confirmed operational with storage, memory, internet, DNS, and dual-adapter connectivity checks.

Item	Validated Result
Source	NGT Academy training material using a provided Google Drive link
Guest name	Kali Linux - NGT
Resources	8 GB RAM, 4 vCPUs, 80 GB virtual disk
Network interfaces	eth0 and eth1 on separate NAT networks
Operating system	Kali GNU/Linux Rolling 2026.1
Kernel	7.0.12+kali-amd64
Architecture	x86-64
Platform	Oracle VirtualBox

Source and Deployment Method

The Kali virtual appliance was obtained through NGT Academy training material rather than directly from the public Kali download page. The source was documented accurately and the appliance was imported as provided.

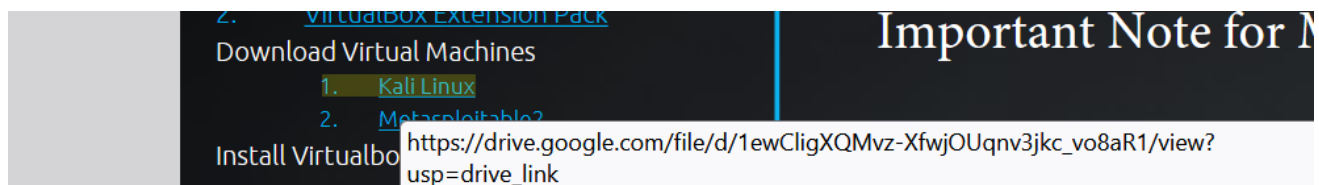


Figure 1. NGT training material providing the Kali Linux virtual machine download link

Virtual Appliance Import

1. Download the NGT-provided Kali appliance to the lab account.
2. Open VirtualBox Manager and choose Import.
3. Select the local OVA file.
4. Review the imported guest configuration before finishing.
5. Confirm the virtual machine appears in the manager and is powered off before network changes.

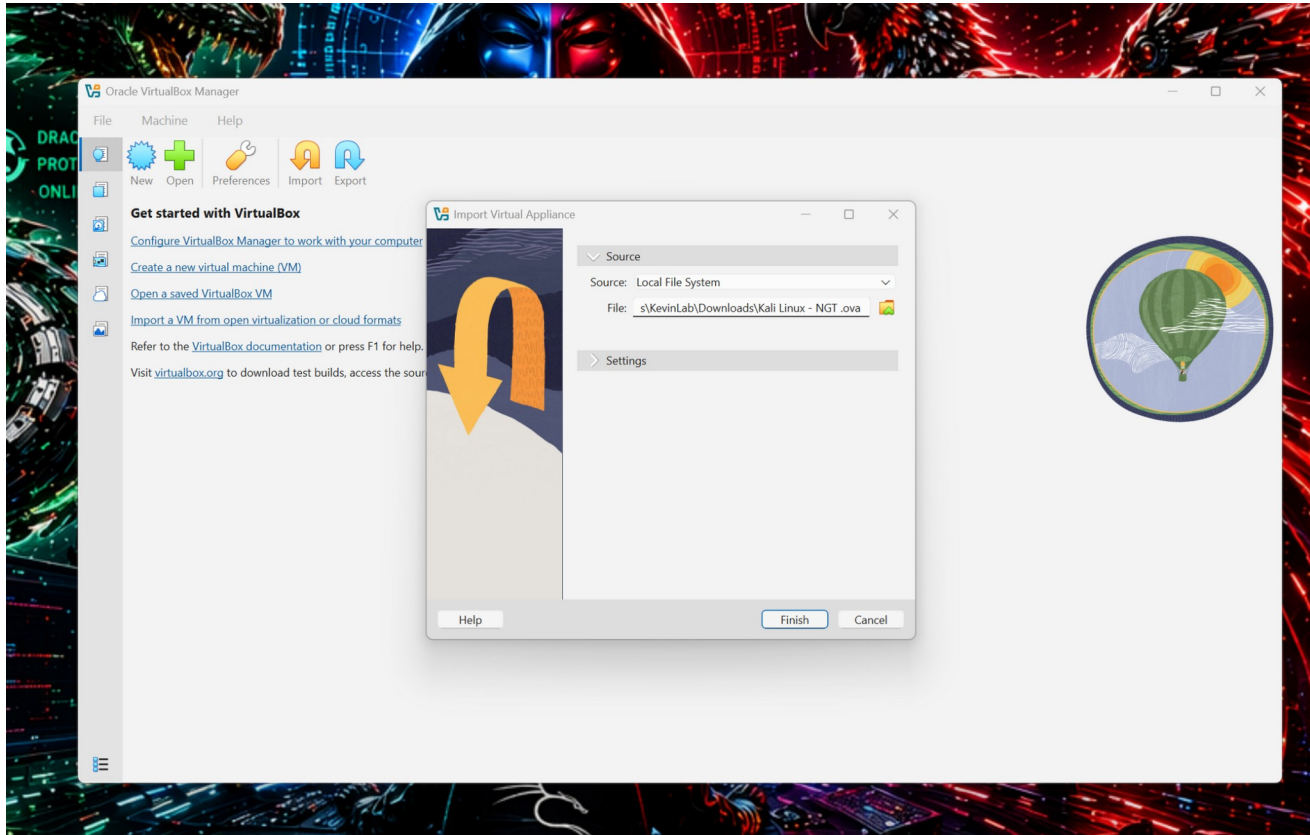


Figure 2. Kali OVA selected from the lab account Downloads folder

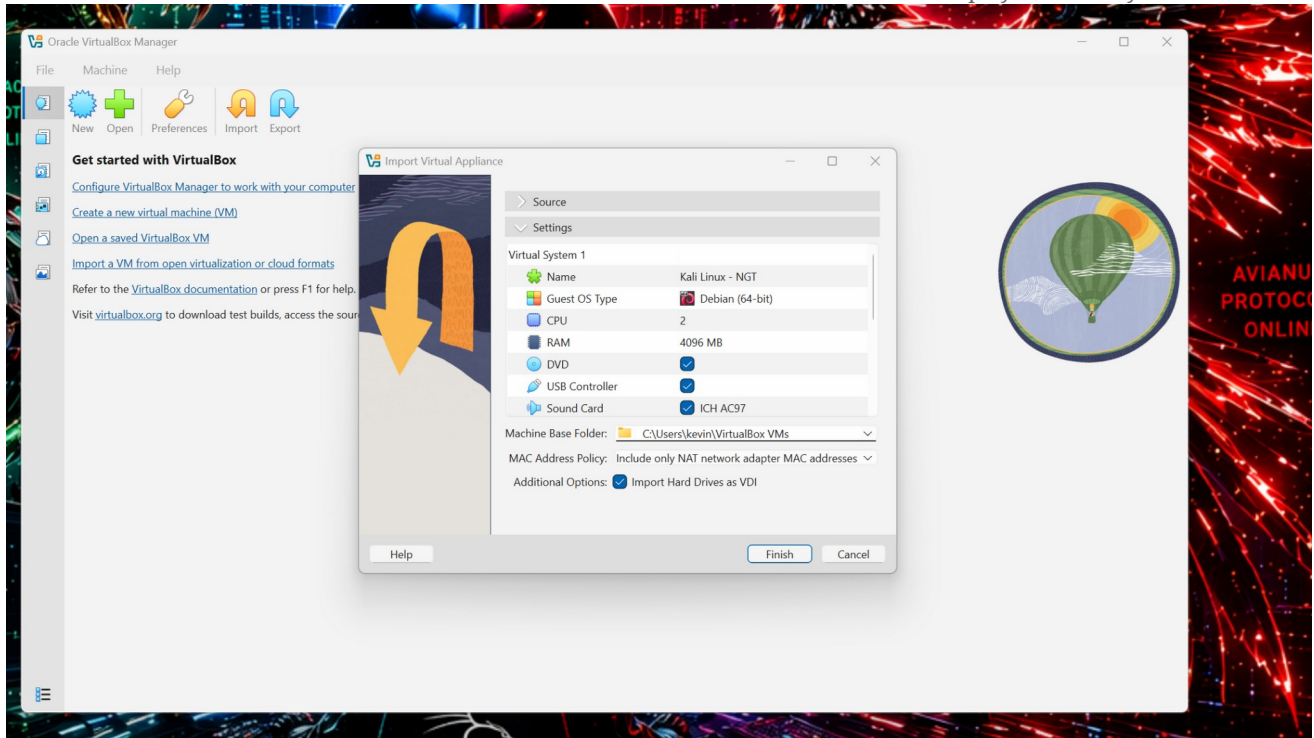


Figure 3. Virtual appliance import summary

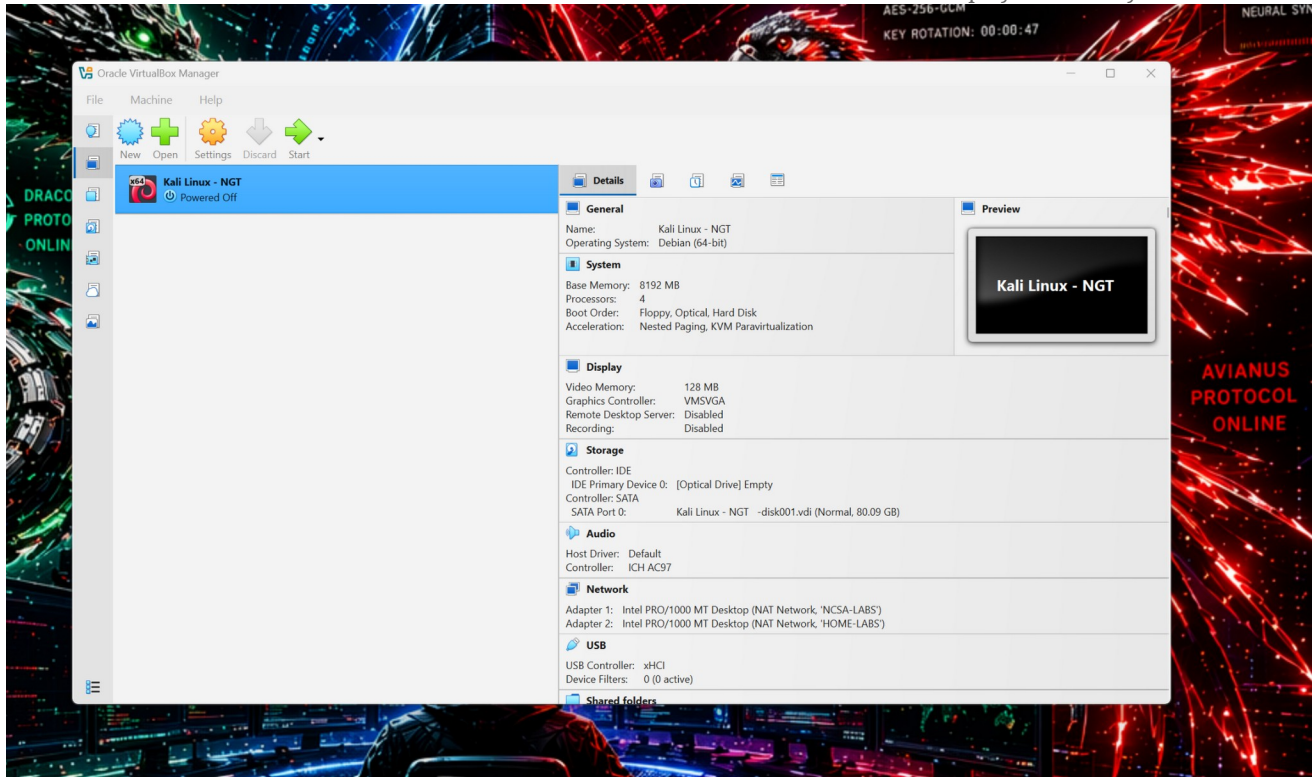
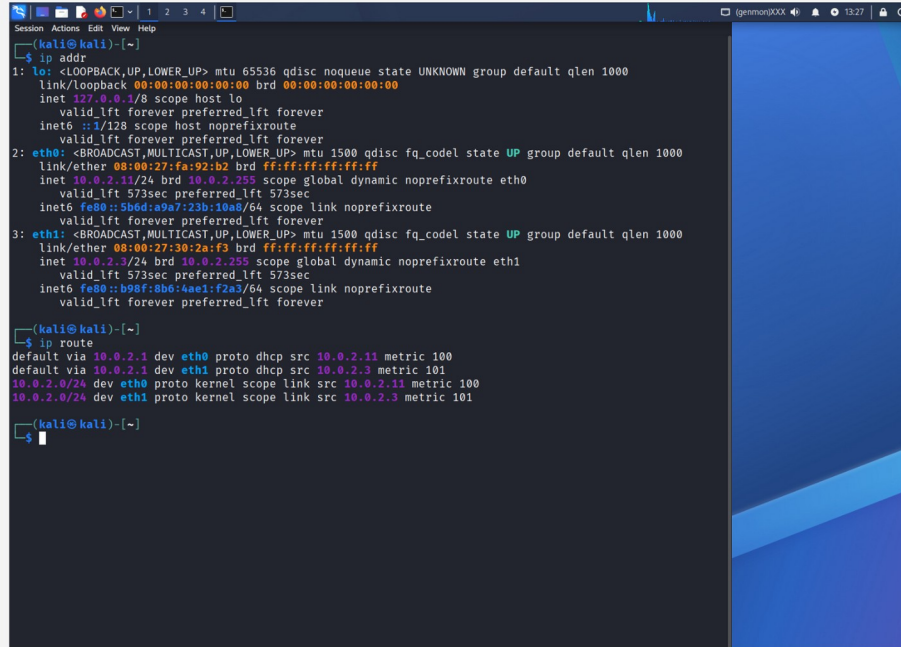


Figure 4. Imported Kali VM configuration

Initial Login and Configuration

The imported appliance booted to the Kali login screen and was accessed with the lab credentials supplied for the training environment. No passwords are reproduced in this documentation.



```
(kali@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:fa:92:b2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.11/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 573sec preferred_lft 573sec
    inet6 fe80::5b6d:a9a7:23b:10a8/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:30:2a:f3 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.3/24 brd 10.0.2.255 scope global dynamic noprefixroute eth1
        valid_lft 573sec preferred_lft 573sec
    inet6 fe80::b98f:8b6:4ae1:f2a3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ ip route
default via 10.0.2.1 dev eth0 proto dhcp src 10.0.2.11 metric 100
default via 10.0.2.1 dev eth1 proto dhcp src 10.0.2.3 metric 101
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.11 metric 100
10.0.2.0/24 dev eth1 proto kernel scope link src 10.0.2.3 metric 101

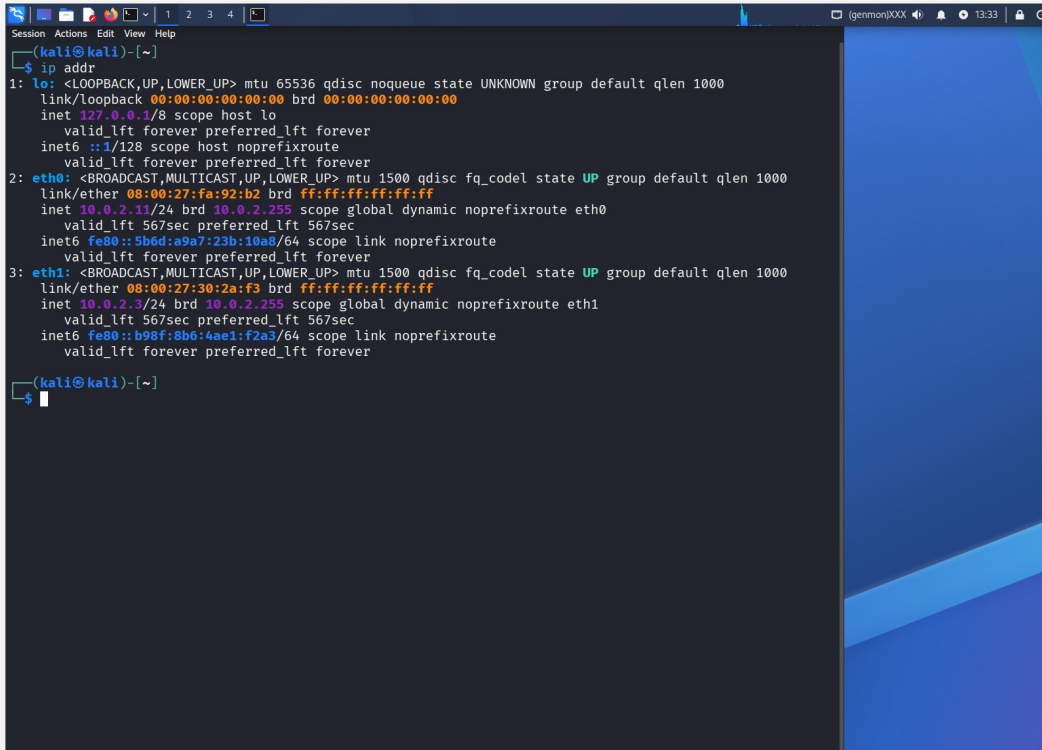
(kali@kali)-[~]
$
```

Figure 5. Kali login screen

Dual-Network Architecture

The Kali VM uses two VirtualBox adapters. The first interface connects to the NCSA-LABS NAT network and the second interface connects to HOME-LABS. Linux interface and routing commands were used to verify both adapters and their assigned addresses.

```
ip -br addr
ip addr
ip route
```



```
(kali)kali-[-]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:fa:92:b2 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.11/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 567sec preferred_lft 567sec
   inet6 fe80::5b6d:a9a7:23b:10a8/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:30:2a:f3 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.3/24 brd 10.0.2.255 scope global dynamic noprefixroute eth1
       valid_lft 567sec preferred_lft 567sec
   inet6 fe80::b98f:8b6:4ae1:f2a3/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
(kali)kali-[-]
$
```

Figure 6. Kali network interfaces after dual-adapter configuration



Figure 7. Kali IP addresses and route table

System Update and Maintenance

Because Kali is a rolling distribution, package indexes and installed software were updated together. Obsolete packages and older kernels were then removed, package caches were cleaned, and the repository state was rechecked.

```
sudo apt update
sudo apt full-upgrade -y
sudo apt autoremove -y
sudo apt autoclean
sudo apt update
```

Maintenance Result

The update workflow completed successfully and the final repository check reported that all packages were up to date.

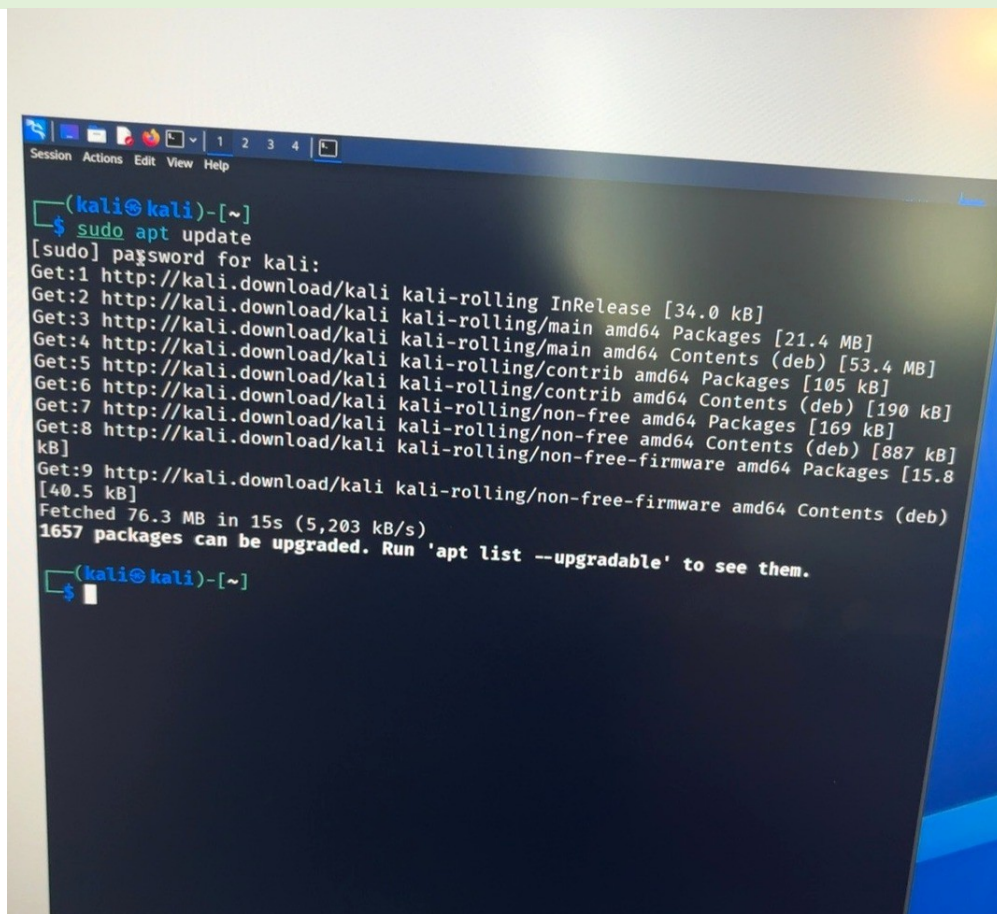
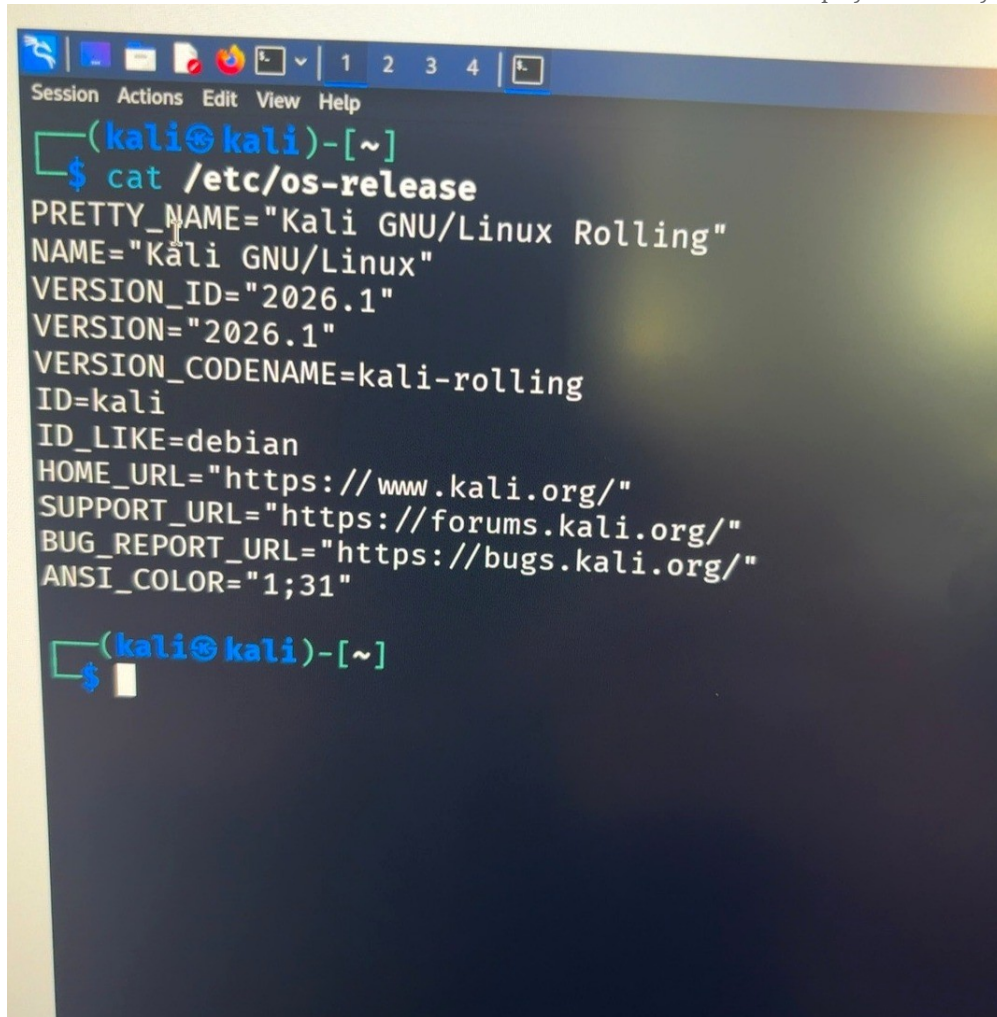


Figure 8. Kali package index refresh showing available upgrades

A terminal window with a dark background and light blue text. The window has a title bar with icons for session, actions, edit, view, and help. The terminal shows the command 'cat /etc/os-release' being executed, which outputs the following information: PRETTY_NAME="Kali GNU/Linux Rolling", NAME="Kali GNU/Linux", VERSION_ID="2026.1", VERSION="2026.1", VERSION_CODENAME=kali-rolling, ID=kali, ID_LIKE=debian, HOME_URL="https://www.kali.org/", SUPPORT_URL="https://forums.kali.org/", BUG_REPORT_URL="https://bugs.kali.org/", and ANSI_COLOR="1;31". The prompt '(kali@kali)-[~]' is visible at the top and bottom of the terminal output.

```
Session Actions Edit View Help
(kali@kali)-[~]
$ cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.1"
VERSION="2026.1"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"

(kali@kali)-[~]
$
```

Figure 9. Operating system verification showing Kali GNU/Linux Rolling 2026.1

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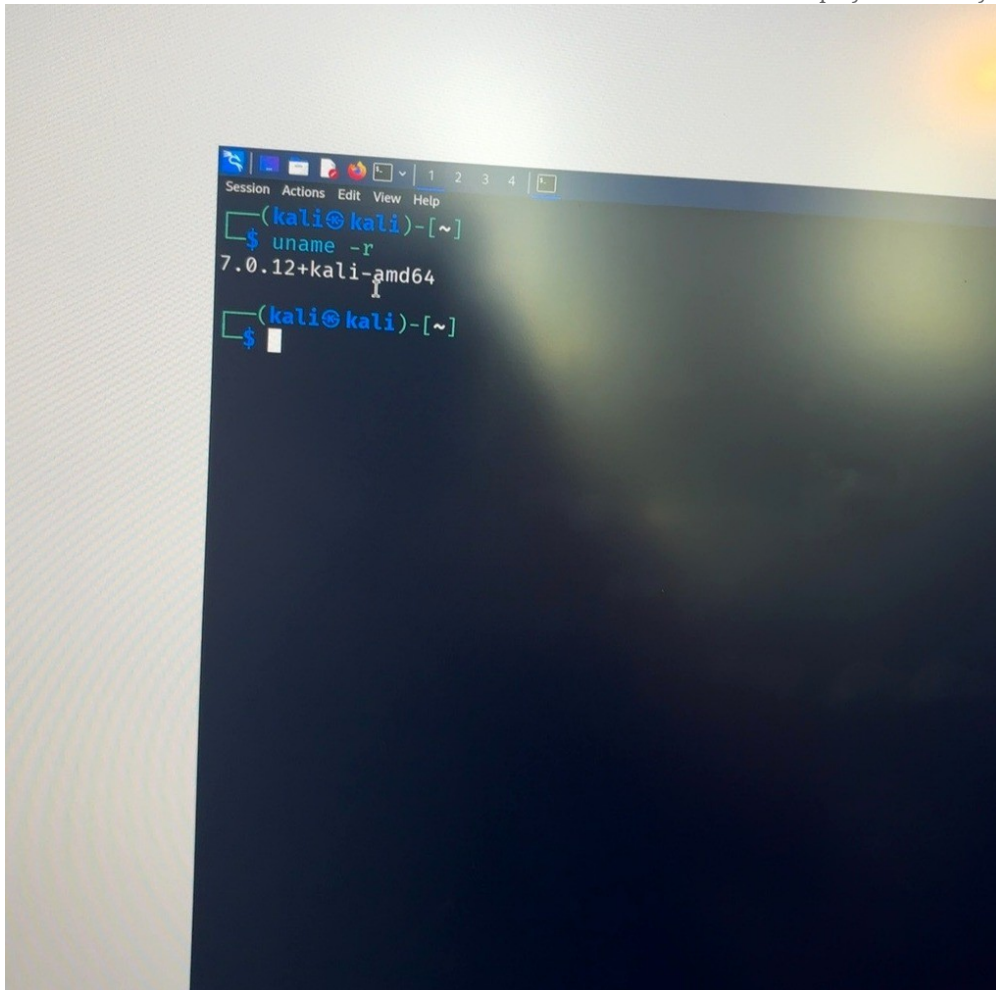


Figure 11. Final package validation showing all packages up to date

Operating System and Kernel Validation

The operating system release file and kernel output were reviewed to confirm that the imported system had been updated successfully.

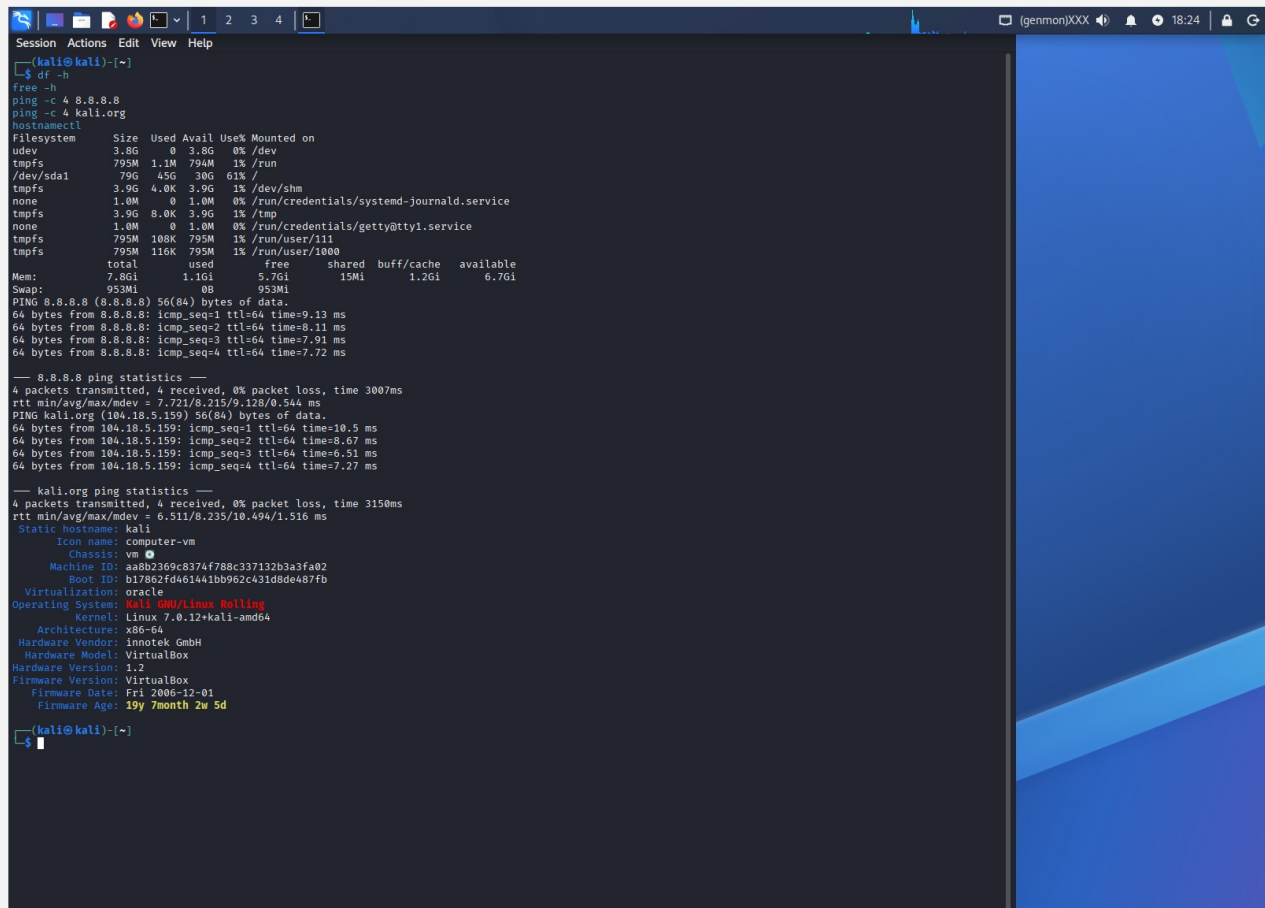
```
cat /etc/os-release
uname -r
hostnamectl
```

Validation	Observed Value
Distribution	Kali GNU/Linux Rolling
Version	2026.1
Kernel	7.0.12+kali-amd64
Architecture	x86-64
Virtualization	Oracle
Hardware model	VirtualBox

Final Operational Validation

A consolidated terminal check verified the guest operating system after updates and cleanup.

```
df -h
free -h
ping -c 4 8.8.8.8
ping -c 4 kali.org
hostnamectl
```



```

(kali@kali)-[~]
$ df -h
$ free -h
$ ping -c 4 8.8.8.8
$ ping -c 4 kali.org
$ hostnamectl
Filesystem      Size  Used Avail Use% Mounted on
udev            3.8G     0  3.8G   0% /dev
tmpfs           795M    1M  794M   1% /run
/dev/sda1       79G   45G   30G  61% /
tmpfs           3.9G   4.0K   3.9G   1% /dev/shm
none            1.0M     0   1.0M   0% /run/credentials/systemd-journald.service
tmpfs           3.9G   8.0K   3.9G   1% /tmp
none            1.0M     0   1.0M   0% /run/credentials/getty@tty1.service
tmpfs           795M  108K  795M   1% /run/user/111
tmpfs           795M  116K  795M   1% /run/user/1000
total           7.8Gi    1.1Gi   5.7Gi    15M    1.2Gi   6.7Gi
Mem:            953Mi    1.1Gi    84Mi    15M    1.2Gi   6.7Gi
Swap:           953Mi     0B    953Mi

PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=9.13 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=8.11 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=7.91 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=7.72 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 7.721/8.215/9.128/0.544 ms
PING kali.org (104.18.5.159) 56(84) bytes of data:
64 bytes from 104.18.5.159: icmp_seq=1 ttl=64 time=10.5 ms
64 bytes from 104.18.5.159: icmp_seq=2 ttl=64 time=8.67 ms
64 bytes from 104.18.5.159: icmp_seq=3 ttl=64 time=6.51 ms
64 bytes from 104.18.5.159: icmp_seq=4 ttl=64 time=7.27 ms

--- kali.org ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3150ms
rtt min/avg/max/mdev = 6.511/8.235/10.494/1.516 ms

$ cat /etc/os-release
Static hostname: kali
Icon name: computer-vm
Chassis: vm
Machine ID: 8a8b2369c8374f780c37122b3a3fa02
Boot ID: b17862fd461441bb962c43108de487fb
Virtualization: oracle
Operating System: Kali GNU/Linux Rolling
Kernel: Linux 7.0.12+kali-amd64
Architecture: x86_64
Hardware Vendor: innotek GmbH
Hardware Model: VirtualBox
Hardware Version: 1.2
Firmware Version: VirtualBox
Firmware Date: Fri 2006-12-01
Firmware Age: 19y 7month 2w 5d

(kali@kali)-[~]
$

```

Figure 12. Final validation of disk, memory, internet, DNS, kernel, architecture, and VirtualBox platform

Check	Result
Disk	79 GB root filesystem, 30 GB available, 61% used
Memory	7.8 GiB total, approximately 6.7 GiB available
Internet	8.8.8.8 reachable with 0% packet loss
DNS	kali.org resolved and responded with 0% packet loss
Hostname	kali
OS / Kernel	Kali Rolling / 7.0.12+kali-amd64

Routine Maintenance Checklist

- Create a snapshot before full upgrades or major tool changes.
- Run apt update and full-upgrade on a planned schedule.
- Review autoremove output before confirming package removal.
- Monitor disk usage with df -h and memory with free -h.
- Verify both virtual interfaces and routing before lab exercises.
- Keep the VM on NAT or isolated lab networks unless a documented exercise requires another mode.
- Export or back up the appliance after major stable milestones.

Interview Talking Points

- Imported an instructor-provided Kali OVA into Oracle VirtualBox and tuned VM resources.
- Configured and validated two independent NAT network interfaces.
- Updated a Kali rolling-release system and removed obsolete packages and kernels.
- Verified operating system release, kernel, architecture, storage, memory, internet, and DNS.
- Documented the source, configuration, troubleshooting, and final validation with screenshot evidence.